

Emanuele Rossi

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Researcher working on machine learning for animal communication, part of a broader interest in leveraging ML to recover structure in complex systems: previously ML on graphs (Temporal Graph Networks, SIGN) at Twitter and Imperial College London, and generative models for biomolecules at VantAI.

Google Scholar metrics: 4,444 citations; h-index 15 (August 2026).

Education

Imperial College London Oct 2020 – Feb 2024

Ph.D. in Computer Science – Supervisor: Prof. Michael Bronstein

- Research focused on **graph machine learning methods** for large real-world systems

University of Cambridge Oct 2018 – Jun 2019

Master in Advanced Computer Science, Distinction (4.0 GPA)

Imperial College London Oct 2015 – Jun 2018

Bachelor in Computer Science, First Class Honors (4.0 GPA)

Experience

Postdoctoral Researcher Rome (remote)

GLADIA Research Group [🔗](#), Sapienza University of Rome – With Prof. E. Rodolà Oct 2025 – Present

- Developing machine learning methods for animal communication.

Machine Learning Researcher NYC (remote)

VantAI Jan 2024 – Aug 2025

- **Co-led development of Neo-1 [🔗](#)**, a state-of-the-art all-atom latent diffusion model for multi-modal structure prediction and de-novo generation of biomolecular complexes, trained on 100+ GPUs.

Machine Learning Researcher London

Twitter Jun 2019 – Feb 2023

- Developed **graph ML models** for Twitter’s social graph and deployed a recommendation model that generated **1.2M additional daily follows**

Machine Learning Researcher London

Fabula AI Mar 2019 – Jun 2019

- Developed graph ML models for fake news detection as part of Fabula AI’s small research team before **Twitter acquired the company [🔗](#)**

Software Engineering Intern California

Google Jul 2017 – Oct 2017

Selected Publications

[Full list](#)

Animal Communication & Audio

- *D. Marincione, D. Crisostomi, R. Dessì, E. Rodolà, and **E. Rossi**. [Model Merging Improves Zero-Shot Generalization in Bioacoustic Foundation Models \[🔗\]\(#\)](#). *NeurIPS Workshop on AI for Animal Communication 2025, Spotlight (top 15%)*.*
- *F. Mustun, C. Semenzin, R. Dessì, P. R. Guerrero, P. Orhan, A. Emanuelli, **E. Rossi** et al. Open-Whistle: A Large-Scale Longitudinal Dataset and Benchmark of Bottlenose Dolphin Vocalizations. *Under Review 2026*.*
- ***E. Rossi** and E. Rodolà. [Communicating Sound Through Natural Language \[🔗\]\(#\)](#). *arXiv 2026*.*

Graph Learning

- ***E. Rossi** et al. [Temporal Graph Networks for Deep Learning on Dynamic Graphs \[🔗\]\(#\)](#). *ICML GRL**

Workshop 2020. **1,700+** citations and **1,000+** GitHub stars [↗](#).

- **E. Rossi***, F. Frasca* et al. SIGN: Scalable Inception Graph Neural Networks [↗](#). ICML GRL Workshop 2020. **600+** citations and used in **production at Airbnb** [↗](#).
- B. Chamberlain, S. Shirobokov, **E. Rossi** et al. Graph Neural Networks for Link Prediction with Subgraph Sketching [↗](#). ICLR 2023, Oral (top 5%). **200+** citations.

Leadership

Co-Founder, LeadTheFuture ([Forbes article](#) [↗](#))

Jun 2018 – Mar 2023

- Built a non-profit mentoring community with more than **200 mentors** and **500 selected mentees**